

Karthik Viswanathan

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I am pursuing a PhD at the University of Amsterdam, focusing on **interpretability** of large language models through **geometric analysis** of their internal representations. My interdisciplinary background began with **competitive programming**, progressed through developing machine learning models as a **surveillance analyst** at Goldman Sachs, and was enriched during my master's in **theoretical physics**. My cross-domain exposure has allowed me to transition smoothly across disciplines, enabling meaningful contributions to both practical implementations and theoretical investigations in scientific research.

Education

University of Amsterdam <i>PhD in Physics, Expected: December 31, 2025</i> Thesis: From Language Models to Cosmic Structures: A Geometric Perspective Advisor: Dr. Jan Pieter van der Schaar	Amsterdam, Netherlands 2021–2025
University of Amsterdam <i>MSc in Physics and Astronomy (Theoretical Physics), cum laude</i> Thesis: Exploring the Spectral Theory/Topological Strings duality Advisor: Dr. Marcel Vonk GPA: 8.7/10	Amsterdam, Netherlands 2019–2021
Indian Institute of Technology Madras <i>B.Tech in Engineering Physics</i> Thesis: Real Space Renormalization and Applications to Machine Learning Advisor: Dr. Ashwin Joy GPA: 8.29/10	Chennai, India 2013–2017

Research Experience

Probing Geometry of Next Token Prediction Using Cumulant Expansion of Softmax Entropy Karthik Viswanathan, Sang Eon Park <i>ICML 2025: Workshop on High-dimensional Learning Dynamics</i>	
Persistent Topological Features in Large Language Models Yuri Gardinazzi*, Karthik Viswanathan*, Giada Panerai, Alessio Ansuini, Alberto Cazzaniga, Matteo Biagetti <i>Proceedings of the 42nd International Conference on Machine Learning</i> *Equal contribution	
The Geometry of Tokens in Internal Representations of Large Language Models Karthik Viswanathan, Yuri Gardinazzi, Giada Panerai, Alberto Cazzaniga, Matteo Biagetti <i>Under review</i>	
Cosmology with persistent homology: A Fisher forecast Jacky Yip, Matteo Biagetti, Alex Cole, Karthik Viswanathan, Gary Shiu <i>Journal of Cosmology and Astroparticle Physics (JCAP), Vol. 2024, Issue 09</i>	
Mentorship for Alignment Research Students (MARS 3.0) Cambridge AI Safety Hub (Summer 2025) <i>Developing methods to elicit more faithful and complete chain of thought in LLMs.</i>	
Workshop: Interpretability in LLMs using Geometric and Statistical methods Karthik Viswanathan, Jan Pieter van der Schaar <i>Organized a two-day workshop with around 40 participants from Amsterdam and AREA Science Park (Trieste).</i>	

Professional Experience

Goldman Sachs <i>Surveillance Analyst</i> Developed ML models for anomaly detection in financial data Manager: Prof. Howard Karloff	Bangalore, India 2017–2019
Goldman Sachs <i>Summer Intern</i> Implemented fast Personalized PageRank in a MapReduce framework Mentor: Dr. Koushik Balasubramanian	Bangalore, India 2016

Awards & Distinctions

Sander Bais Prize for Academic Merit <i>Institute for Theoretical Physics Amsterdam</i> For exceptional academic performance in the master's program	Amsterdam, Netherlands 2020
Honorable Mention <i>ACM ICPC World Finals</i> Represented India in the international collegiate programming contest.	Marrakech, Morocco 2015
Participant <i>International Olympiad in Informatics Training Camp</i> Selected after ranking in top 25 in the Indian National Olympiad in Informatics	Bangalore, India 2013

Organization & Mentoring

AREA Science Park, Trieste <i>Master Student Supervisor</i> Student: Giada Panerai Thesis: Zig-Zag Persistence in Neural Networks Representations	2024
University of Amsterdam <i>Master Student Mentor</i> Student: Sibilla Bouche Thesis: The persistence of non-Gaussian features: a Neural Ratio Estimation approach	2023
University of Amsterdam <i>Master Student Supervisor</i> Student: Sliem el Ela Thesis: From Primordials to Persistence: A Dual Exploration of Multiparameter Topology and Cosmic Origins	2023
University of Amsterdam <i>Journal Club Organizer</i> Cosmology Journal Club	2022

Technical Skills

Programming: PyTorch, TensorFlow, Hadoop, MapReduce, Mathematica
HPC Resources: LUMI consortium, Snellius supercomputer

Teaching

University of Amsterdam

Teaching Assistant

Courses: Topological Data Analysis: a Physics Perspective, Advanced Cosmology, ML for Physics and Astronomy, QFT in Curved Spacetimes, General Relativity

MSc Physics Program

2020–2024

Research Visits

2024: AREA Science Park, Trieste, Italy

September 16 – October 4

2024: AREA Science Park, Trieste, Italy

February 15 – April 27

References

Dr. Jan Pieter van der Schaar

University of Amsterdam

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Dr. Matteo Biagetti

AREA Science Park, Trieste

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Dr. Magnus Botnan

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Dr. Alberto Cazzaniga

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